

Modelling the Structure and Growth Mode of Bamboo-Like Carbon Nanotubes (BCNTs)



Supervised by:

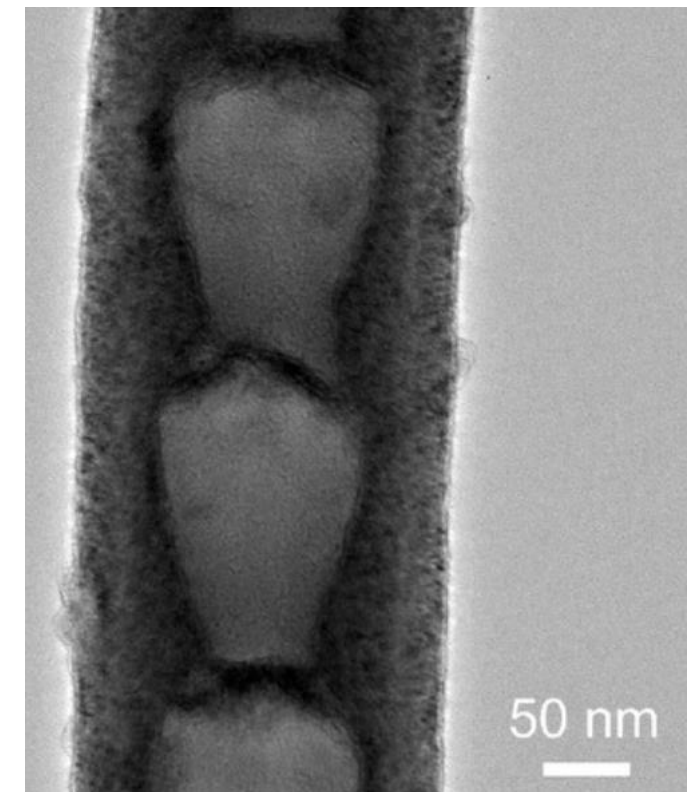
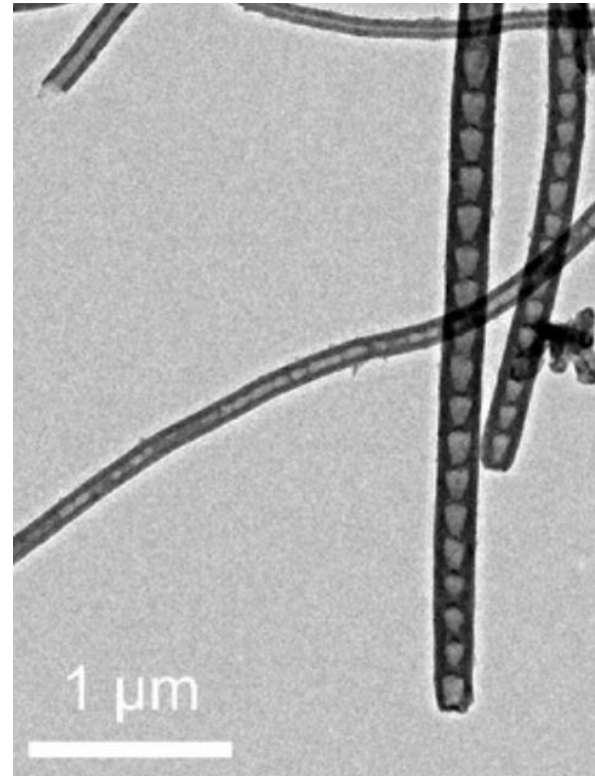
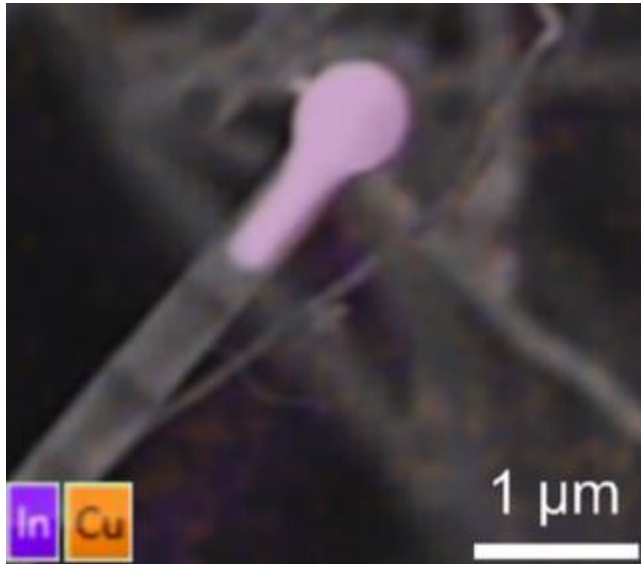
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Two Models:

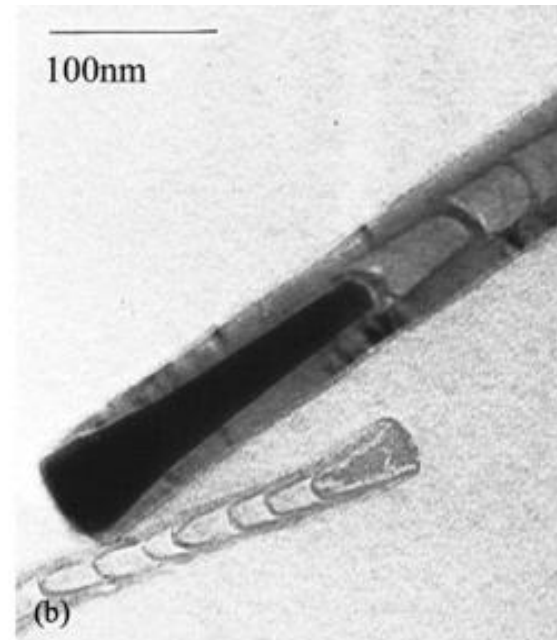
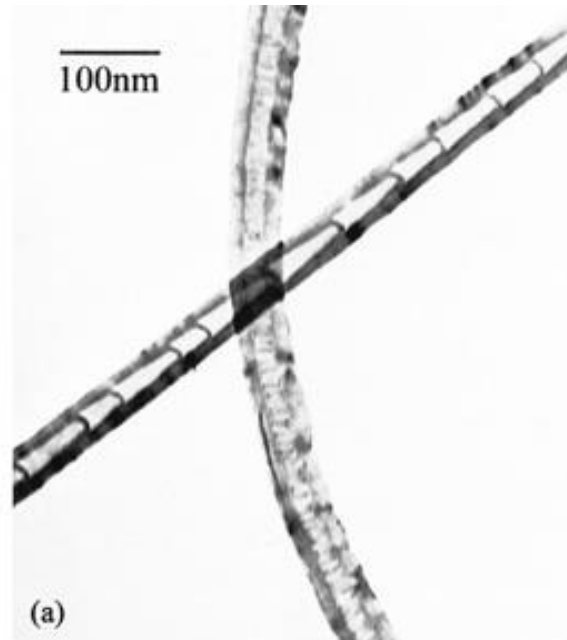
Based on diaphragm formation and detachment, relative to CNT wall growth



- Model 1: Detach First Shoot Second ~ Jellyfish Model
- Model 2: Steady Growth Periodic Detachment ~ Rocket Stage Model

Growth Modes of Bamboo type Carbon Nanotubes: Literature Review

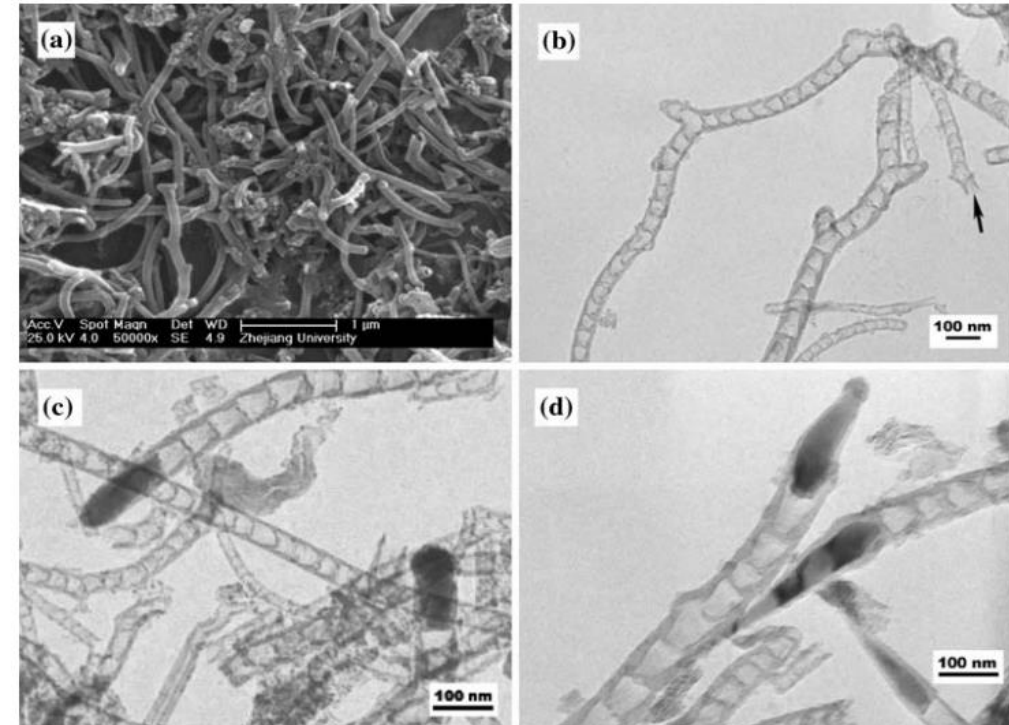
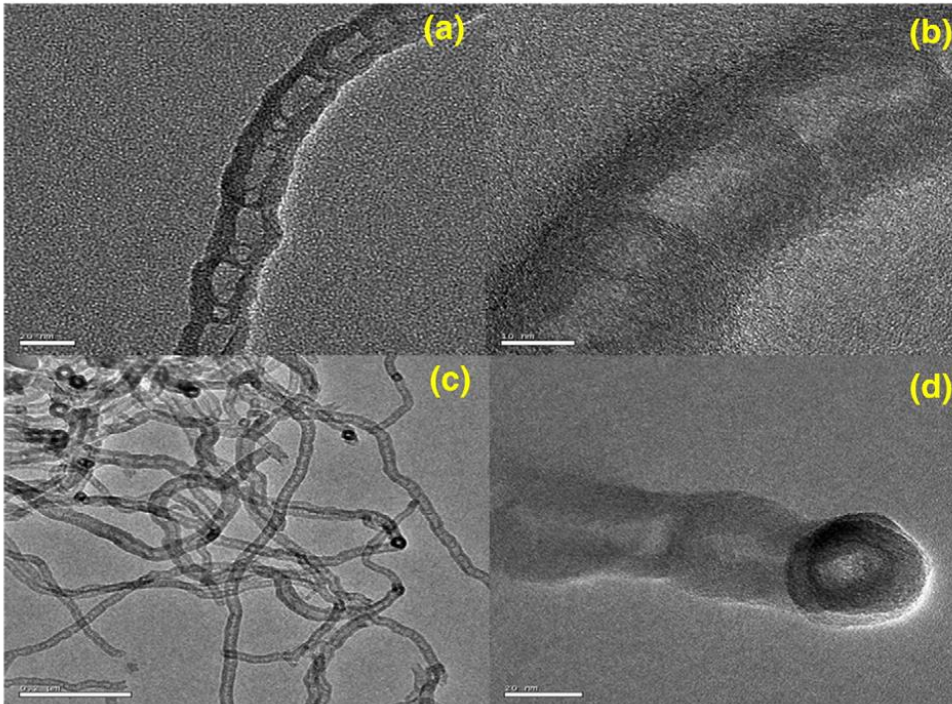
- Papers suggest an intermittent growth model: formation of graphite layers until accumulated stress "propel" catalyst/nanotube.
- Continual growth of old nanotube segment and re-nucleation of new diaphragm.
- Nanotube re-nucleation is theorized to be the rate determining step.



Growth Modes of Bamboo type Carbon Nanotubes: Literature Review

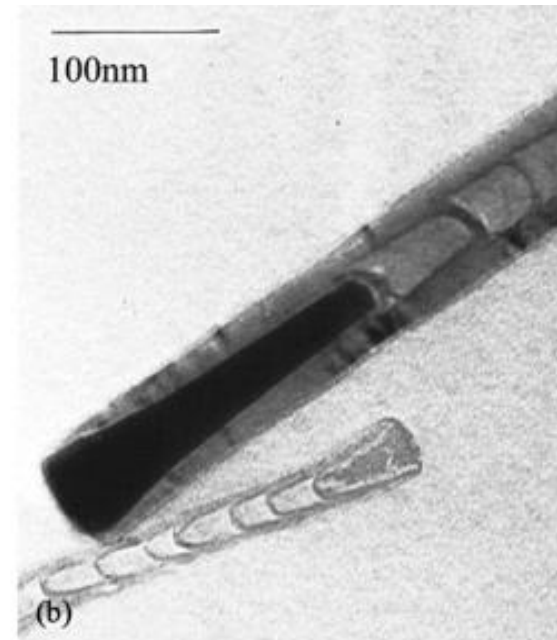
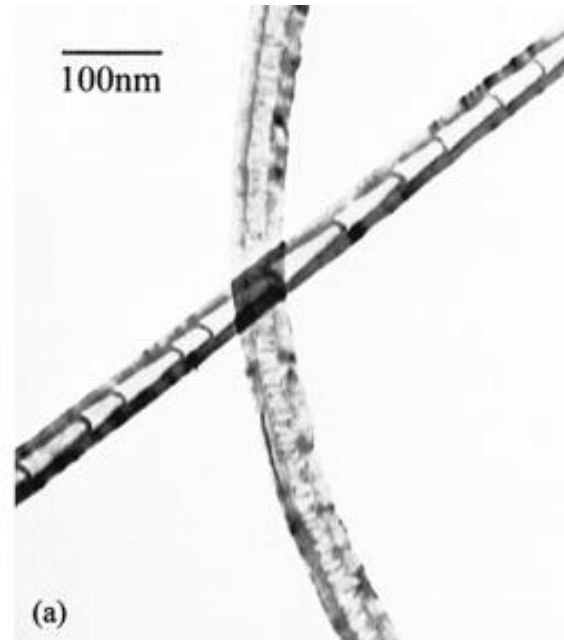


- Factors hypothesized to affect BCNT growth:
 - Dynamic shaping of catalyst nanoparticle
 - Bulk diffusion of carbon
 - Stress accumulation
 - Wettability
 - Adhesive forces between catalyst and CNT



Growth Modes of Bamboo type Carbon Nanotubes: Literature Review

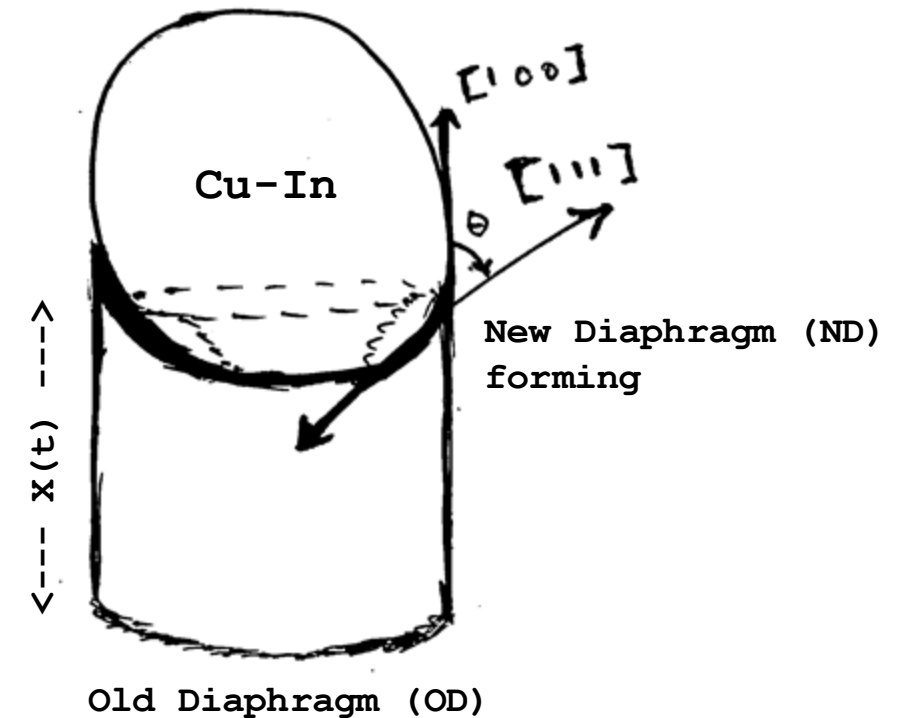
- Shortfalls:
 - Most of the literature focuses on base growth
 - No experimental or simulation-based validation of suggested growth modes
 - All considered catalysts in solid or quasi-liquid state, no studies on molten and/or alloy systems
 - Lack of a clear mechanism to explain effects of temperature and gas pressure



Two Models:

Based on diaphragm formation and detachment, relative to CNT wall growth

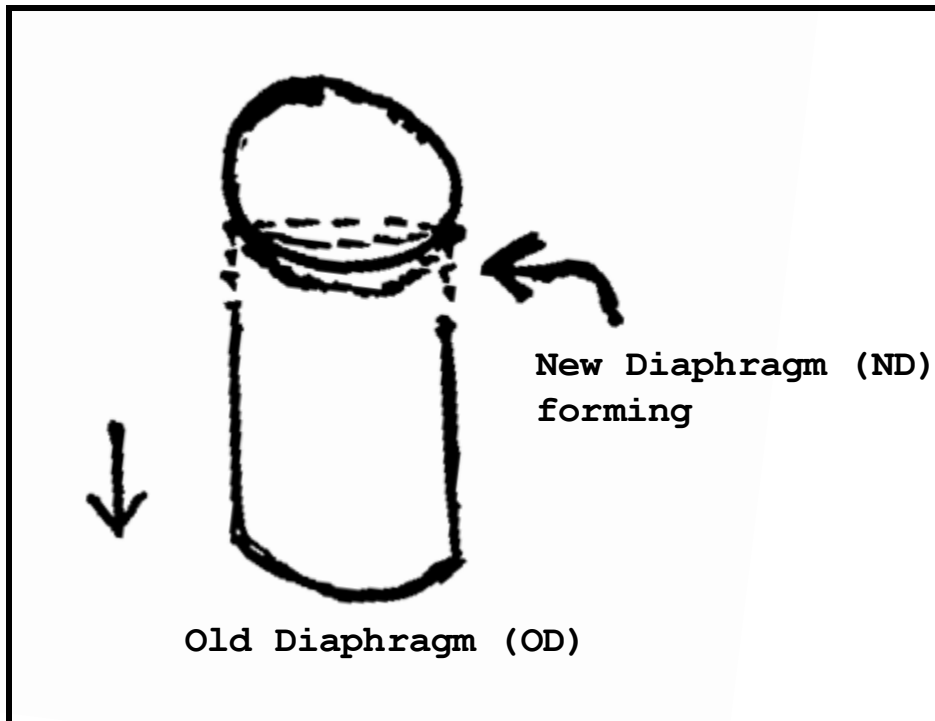
- Function of Knot Wall Growth [length]:
 $x(t) \rightarrow$ Periodic, continuously increasing within the period
- Function of ND Atomic Precipitation
 [carbon atoms/unit surface area of catalyst]:
 $n_p(t) \rightarrow$ Periodic, continuous increasing within the period
- Function of Mechanical Stress applied on Catalyst Surface due to ND formation
 [force/unit surface area of catalyst]:
 $\sigma(t) \rightarrow$ Periodic, continuous increasing within the period
- Reasons for growth and precipitation:
 - Supersaturation
 - Facets with favourable crystalline geometry
 ($\sim$ morphology of the catalyst)
- Period of a Knot Cycle: $t = T$
- Length of a Knot: $x = L$
- Detachment Concentration: $n_p = n_d$
- Detachment Strength: $\sigma = \sigma_d$



Model 1: Detach First Shoot Second (Jellyfish)

Stage 1: Initial Growth

$t = 0$ to t_1

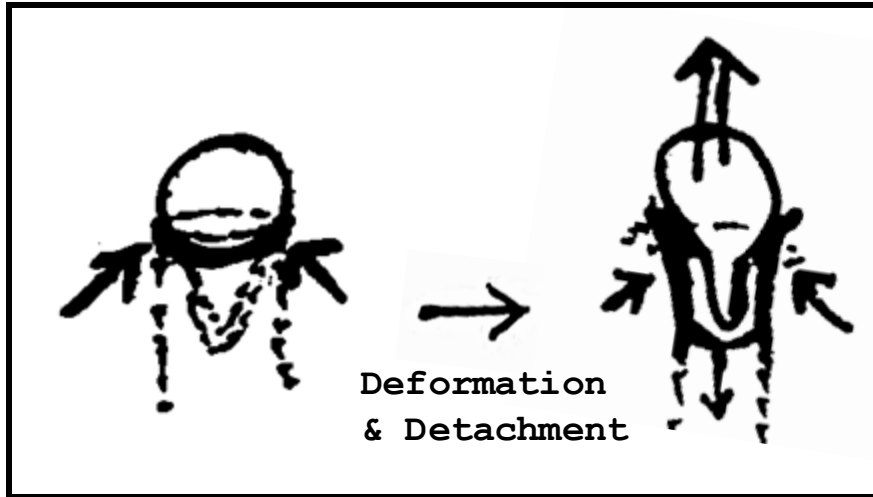


- Nascent Diaphragm (ND) forming. CNT growth is very slow here.
- Reasons:
 - Reduction of surface anisotropy
 - Facets prefer uniform ND formation over CNT Growth

Model 1: Detach First Shoot Second (Jellyfish)

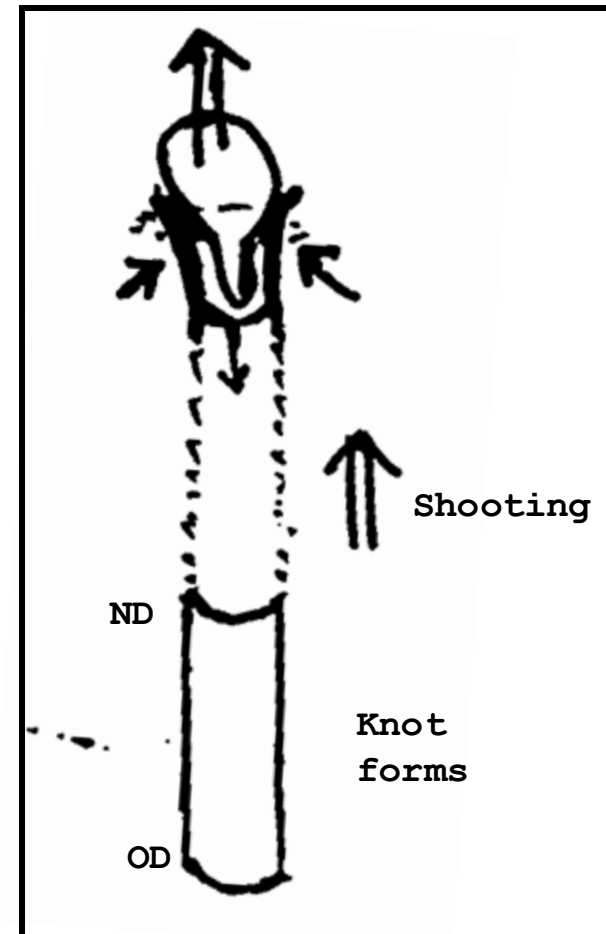
Stage 2: Knot Formation Step

$$t = t_1 \text{ to } t_1 + \Delta t_D + \Delta t_S$$



- Stress Accumulation at ND
- Surface energy of catalyst surface near ND increases
- Leads to Catalyst Deformation (Head and Tail)
- ND detaches
- Detachment occurs due to:
 - Said accumulation of stress
 - Net Bond energy release
 - This energy overcomes strain and surface energy penalties

- Leads to sudden growth (shooting)
- Accumulated Stress > Work Adhesion
- ND is left behind



Stage 3: Catalyst Reset

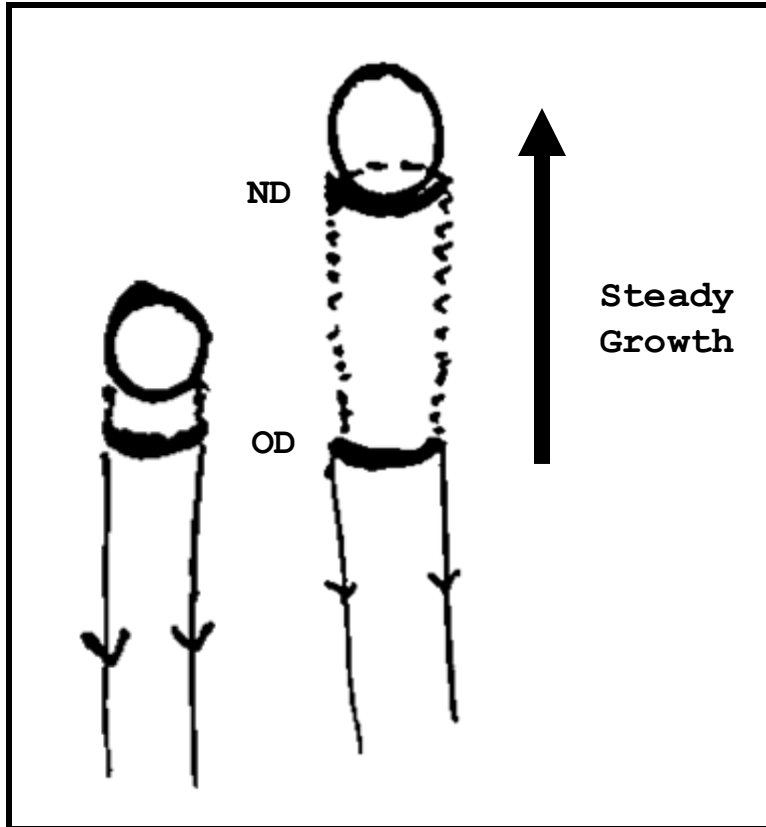
$$t = t_1 + \Delta t \text{ to } T$$

- Surface energy release and minimisation causes catalyst tail to contract.
- Back to Stage One.

Model 2: Steady Growth Periodic Detachment (Rocket Stages)

Stage 1: Initial Growth

$t = 0$ to t_1

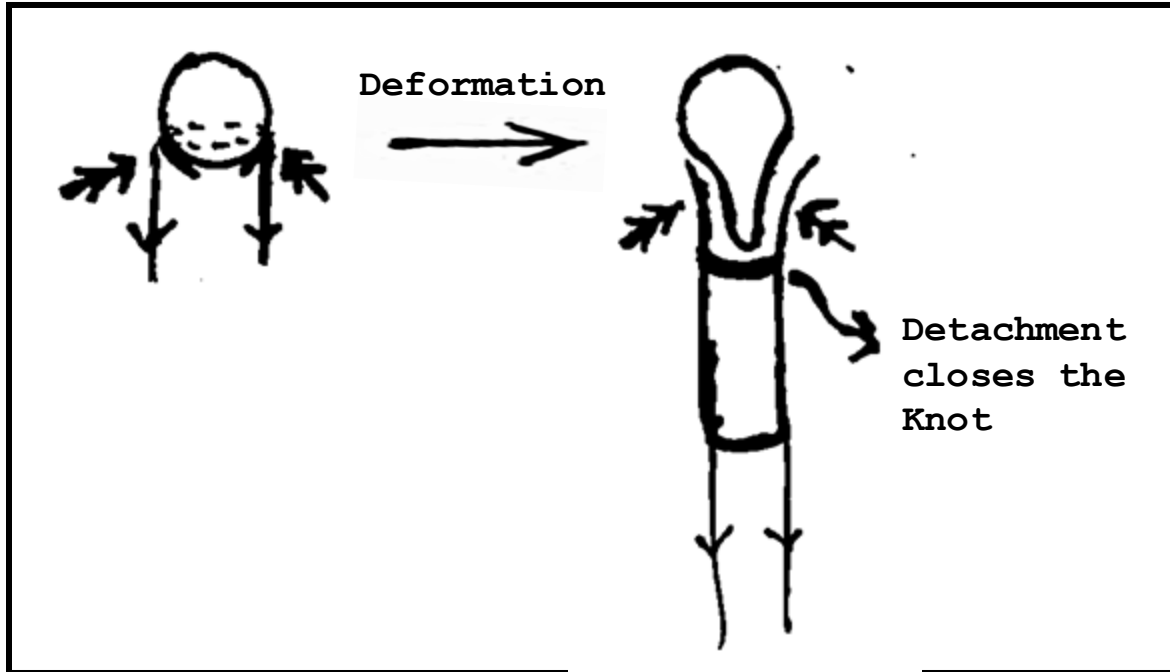


- Growth is steady and continuous.
- OD left behind continuously.
- ND formation occurs continuously.
- No Shooting as Accumulated Stress < Work Adhesion
- Reasons:
 - Reduction of surface anisotropy.
 - Growth site precipitation is not enough
 - Facets also prefer ND precipitation.

Model 2: Steady Growth Periodic Detachment (Rocket Stages)

Stage 2: Knot Closing

$t = t_1$ to $t_1 + \Delta t_D$



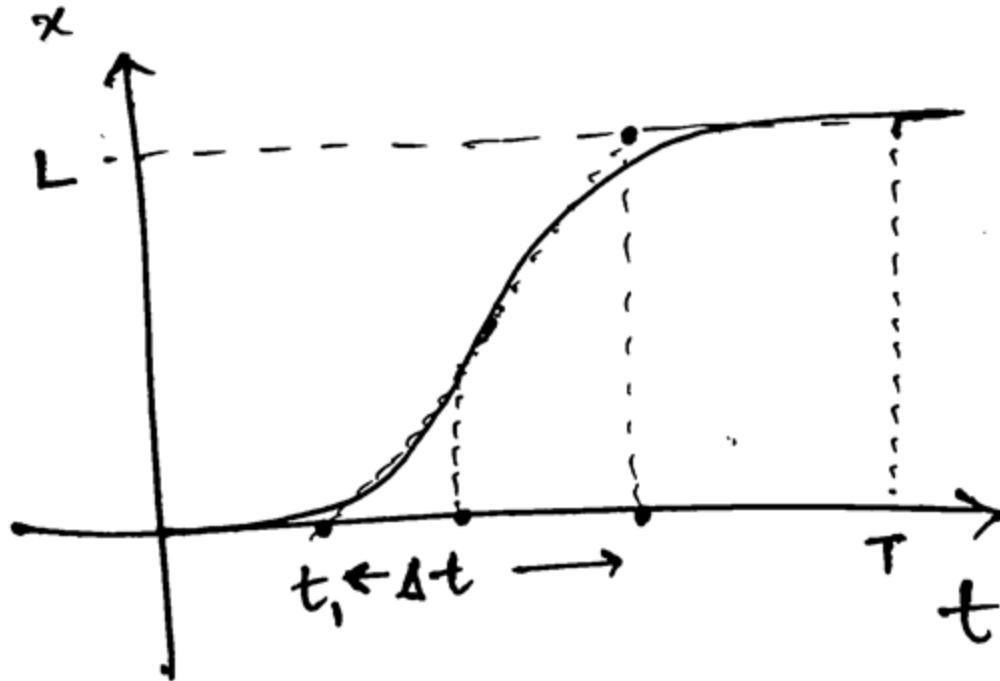
- Stress Accumulation at ND
- Surface energy of catalyst surface near ND increases.
- Leads to Catalyst Deformation (Head and Tail)
- ND detaches
- Detachment occurs due to:
 - Said accumulation of stress
 - Net Bond energy release
 - This energy overcomes strain and surface energy penalties.

Stage 3: Catalyst Reset

$t = t_1 + \Delta t$ to T

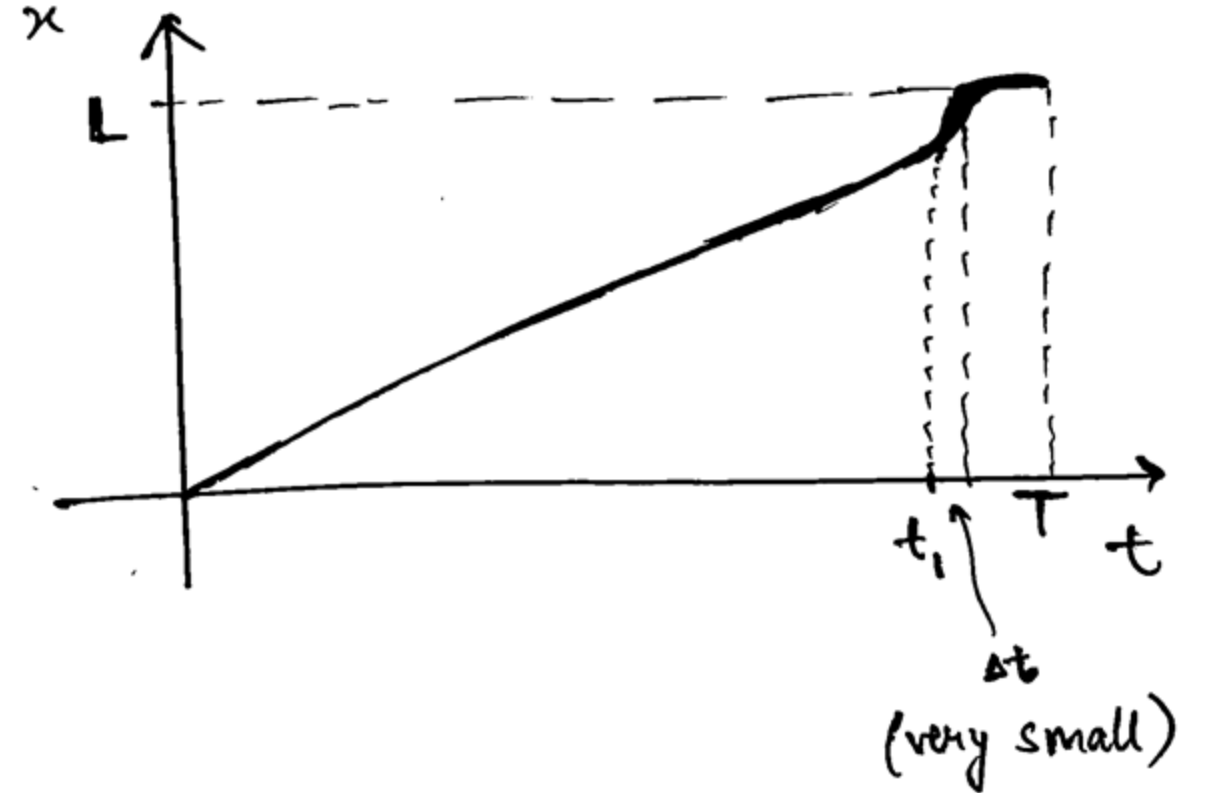
- Surface energy release and minimisation causes catalyst tail to contract.
- Back to Stage One.

Evolution of the Knot



$$\Delta t = \Delta t_D + \Delta t_s$$

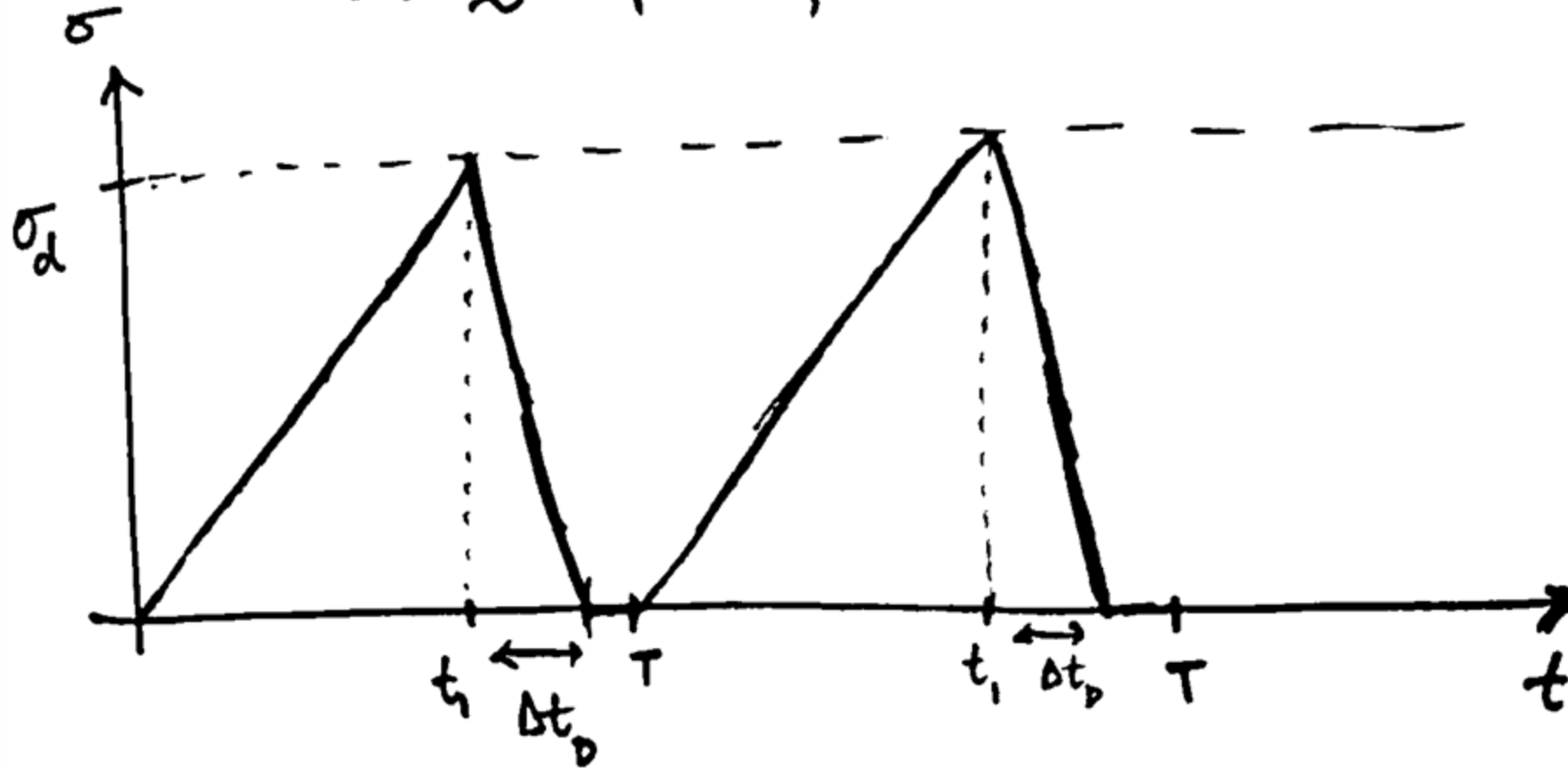
MODEL 1 - Jellyfish



MODEL 2 - Rocket Stages

Detachment Limits

$\sigma(t) \propto \eta_p(t)$, assuming linearity



Detachment Criterion



- Hypothesis: Precipitated graphitic ND behaves like a thin film
- The necessary condition for detachment:
Accumulated Mechanical Stress > Work of Adhesion b/w ND and catalyst
- Mathematical Model:

$$\frac{\sigma^2 h(t)(1 - \nu)}{2E} \geq \gamma(1 + \cos\theta_c)$$

σ : Compressive Stress

$h(t)$: Thickness (time-dependent)

$E/(1-\nu)$: Biaxial Modulus

ν : Poisson's Ratio

γ : Catalyst Surface Energy

θ_c : Contact Angle

Physical Insight



- Wettability becomes the controlling factor.
- Liquid Cu is a poor wetter of graphite, which causes the energy barrier for "shooting" to be low.
- This causes energy to leak continuously due to the negligible work of adhesion. This does not allow accumulation of mechanical stress.
- The percentage of Indium in the catalyst, which shows moderate wetting in graphite, becomes key. It can change the contact angle, and hence the energy barrier.

Thermodynamic Control

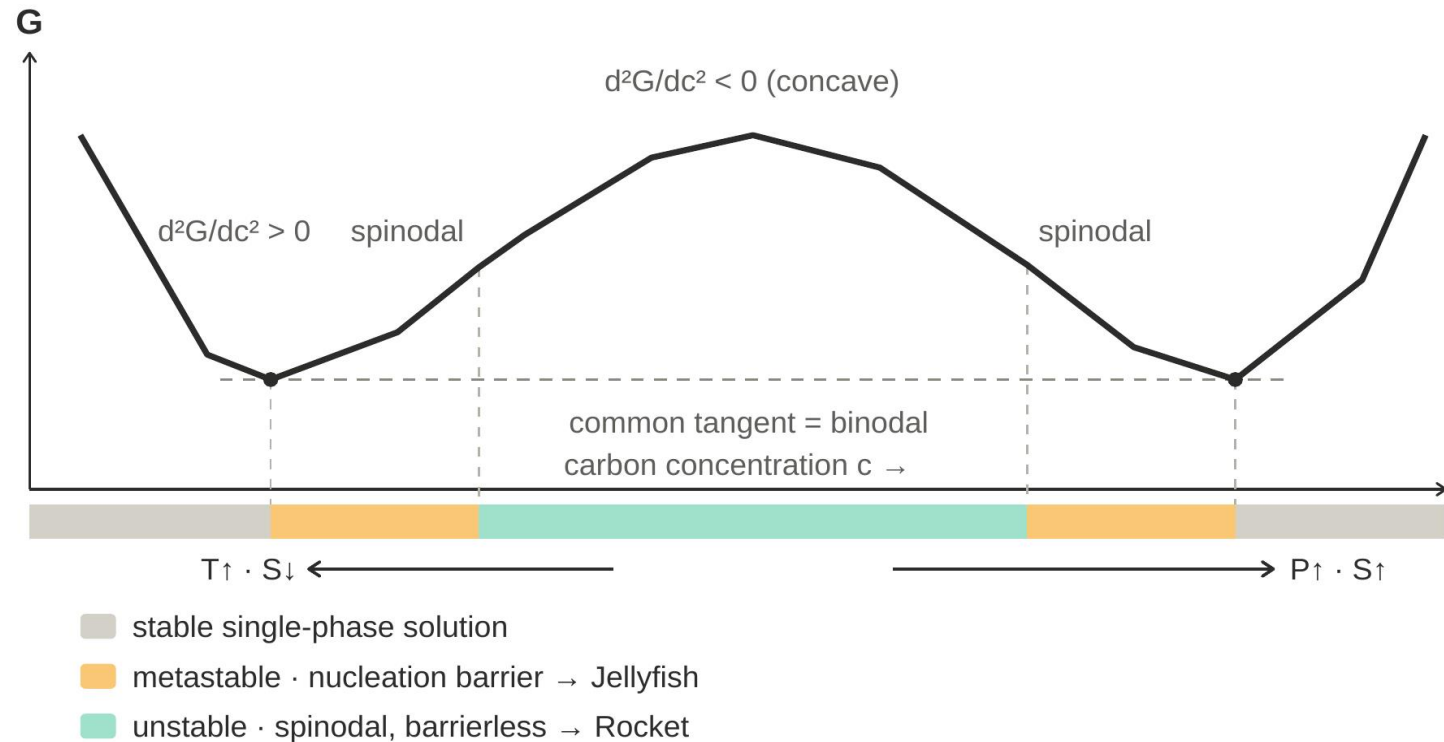


- Gradient of chemical potential of carbon ($\mu_c = \partial G / \partial C$) is the thermodynamic driving force.

- Curvature of the free-energy surface decides the growth mode:

- $\partial^2 G / \partial c^2 > 0$ (metastable): There exists a nucleation barrier, hence carbon precipitates in discrete bursts $\Rightarrow$ Jellyfish Model

- $\partial^2 G / \partial c^2 < 0$ (spinodal): No nucleation barrier, continuous precipitation of carbon $\Rightarrow$ Rocket Model



Kinetic Control

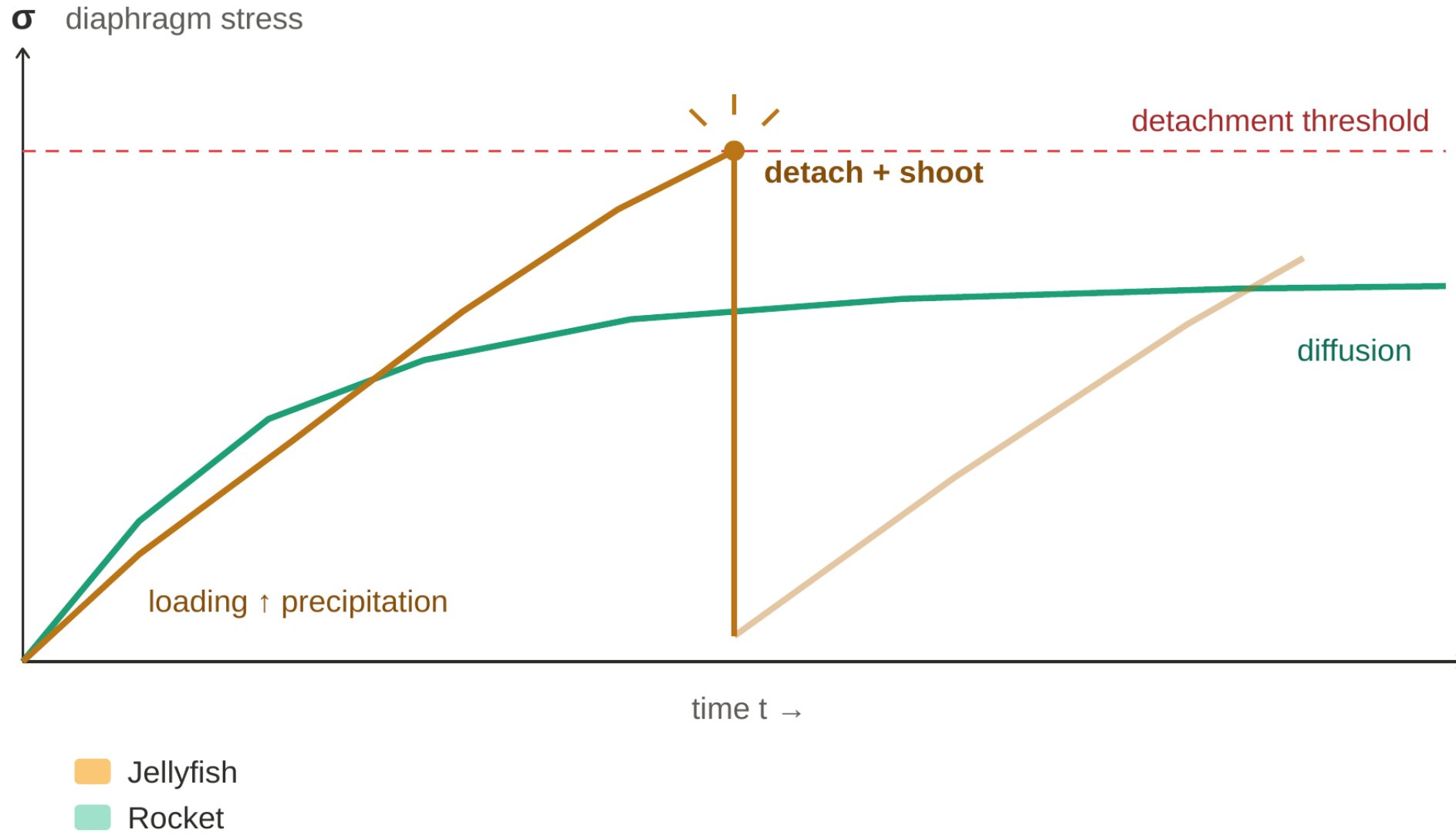


- There is a continuous build-up of energy in the precipitated carbon layers.
- There are two exit pathways for this energy. Selectivity of this pathway is what decides the growth mode.
- We define a dimensionless number:

$D = \text{Rate of Energy Supply} / \text{Rate of Energy Release}$

- The energy is supplied due to supersaturation and subsequent precipitation of carbon (Formation of ND)
- The energy is released by surface diffusion of the precipitated carbon towards the ND edges, aiding tube growth
- If $D \gg 1$: The energy release pathway is not sufficient, which causes stress accumulation. Hence, the systems looks for a mechanical alternative ("shooting") $\Rightarrow$ Jellyfish Growth
Otherwise, it proceeds with Rocket Growth

Kinetic Control - Continued



Key Differences between the models

Aspect	Jellyfish Model	Rocket Stages Model
Growth Mode (Major Contributor)	Shooting Action	Steady Growth
Growth Rate Determining Step	Detachment of New Diaphragm	Carbon Transport Limited
Growth Driving Force	Supersaturation; Stress Release on Detachment triggering shooting action.	Supersaturation; Facets precipitating carbon at the growth site continuously.

- Discussion on Free Energy Thermodynamics?
- Chemical Potential Gradient?



Effect of increasing Precursor Gas Pressure

- Literature indicates increase of ND density
- Possible Reasons:
 - Higher chemical potential gradient
 - Longer Initial Growth
 - Higher Steady Wall Growth Rate tendency
- Possible shift from Jellyfish mode towards Rocket Stage mode.



Effect of increasing Temperature

- Literature indicates an inverse effect of increasing gas pressure.
- Possible Reasons:
 - Higher surface restructuring rate
 - Lower Detachment Limits
 - Faster Initial Growth
 - Possible lower density of ND
- Possible shift from Rocket Stage mode towards Jellyfish mode.